An Integrated Planning Framework for Different Types of PEV Charging Facilities in Urban Area

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***Abstract*—To build a properly planned infrastructure for plug-in electric vehicle (PEV), charging will bolster their market acceptance. Different types of PEV charging facilities for private PEVs, including public charging spots deployed in public parking lots (PLCSs) and roadside fast-charging stations (FCSs), are substitutes for each other. This paper proposes an integrated planning framework for them in an urban area from the perspective of a social planner. The planning objective is to minimize the social costs of the whole PEV charging system. The proposed framework decouples the planning for different types of charging facilities. The spatial and temporal charging demands for FCSs are generated by a charging demand forecasting method, when the quantities of different types of PLCSs are given. The optimal siting and sizing problem of FCSs is solved by Voronoi diagram together with particle swarm optimization algorithm. By traversing the quantities of different types of PLCSs, the optimal planning results are obtained. The effectiveness of the proposed framework is verified via a case study of a real-urban area in China. The substitution effect between different types of charging facilities is studied. The impacts of the ambient temperature, the private charging spot possession rate, and the service level of PLCSs on the planning results are also assessed.**

***Index Terms*—Charging facility, charging load forecasting, integrated planning, particle swarm optimization (PSO) algorithm, plug-in electric vehicles (PEVs), Voronoi diagram.**

NOMENCLATURE

Definitions

HCS1/2	Home charging spot deployed in residential private/public parking lots.
PCS1/2	Public charging spot deployed in public parking lots for employees/the public.
FCS	Fast-charging station deployed on the roadside.
HCS	HCS1 and HCS2.
PCS	PCS1 and PCS2.

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Indices/Sets

i_F	Index of FCSs.
n_{i_F}	Index of the charging spots in FCS i_F.
$n_{H/P}$	Index of the HCS2/PCS charging spots.
t/T	Index/set of time intervals in a day.
j	Index of plug-in electric vehicles (PEVs).
J_{i_F}	Set of PEVs which need to be recharged and whose nearest FCS is i_F in a single day.
s/S	Index/set of day types, divided by ambient temperature.

Parameters

Δt	Duration of each sub-hourly interval, 15 min.
r	Discount rate.
$m_{H/P/F}$	Service lives of the HCS/PCS/FCS, in year.
D_s	Number of s type days in one year.
$C_C^{H/P/F}$	Investment cost per spot for the HCS/PCS/FCS, in \$.
C_G	Grid reinforcement investment cost per unit capacity, in \$/kVA.
C_L	Land cost per square meter, in \$/m^2.
F_L^F	Floor area per FCS spot, in m^2.
C_{IO}^F	Other investment costs of each FCS such as building costs, road improvement costs etc, in \$.
$\mathrm{EC}_{H/P/F}$	Per-unit electricity cost in the HCS/PCS/FCS, in \$/kWh.
$\mathrm{TC}_{\mathrm{dr}}$	Time cost per hour for driving a PEV to a FCS, in \$/h.
$\mathrm{TC}_{\mathrm{ch}}$	Time cost per hour for waiting a PEV to get recharged in a FCS, in \$/h.
v	Average driving speed of PEV j, in km/h.
B_j	Battery size of PEV j, in kWh.
E_j	Energy consumption per kilometer of PEV j, in kWh/km.
η	Average charging efficiency.
$P^{H/P/F}$	Rated charging power of HCS/PCS/FCS charging spot, in kW.

Variables

SoC_j^d	Expected departure SoC of PEV j.
SoC_j^a	Initial SoC of PEV j when it needs to get recharged.
$d_{i_F j}$	Distance between the PEV j which needs to be recharged and its nearest FCS i_F, in km.

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T_j^{w}	Waiting time of PEV j before getting recharged, in h.
$P_{n_{H/P/i_F}}(t)$	Charging power of spot $n_H/n_P/n_{i_F}$ at time t, in kW.
I_F	Number of the FCS.
$N_{H/P}$	Number of the HCS2/PCS charging spots.
N_{i_F}	Number of charging spots in FCS i_F.
l_{i_F}	Location of FCS i_F, $l_{i_F} \subset \mathbb{R}^2$.
$\mathrm{SA}^{H/P}$	Service abilities of the HCS/PCS.
$C_I^{H/P/F}$	Equivalent annual investment costs of the HCS/PCS/FCS, in \$.
$C_{\mathrm{O\&M}}^{H/P/F}$	Annual operation and maintenance costs of the HCS/PCS/FCS, in \$.
$C_E^{H/P/F}$	Annual electricity costs in the HCS/PCS/FCS, in \$.
$C_{E_{\mathrm{ex}}}^F$	Extra electricity costs for driving PEVs to their nearest charging facilities, in \$.
C_T^F	Annual time costs related to the FCS, in \$.
C_I	Total equivalent annual investment costs, in \$.
$C_{\mathrm{O\&M}}$	Total annual operation and maintenance costs, in \$.
C_E	Total annual electricity costs, in \$.
C_T	Total annual time costs, in \$.

I. Introduction

As a cleaner method of transportation with less emission and energy consumption, PEVs have drawn much attention around the world [1]–[5]. Though governments and auto companies have taken great efforts to promote the development of PEVs, there is still a large gap between what was expected since 2008 across the world [6]. Since the inconvenience to get recharged is one of the major hurdles for the promotion of PEVs, to build a properly planned infrastructure for PEV charging will bolster their market acceptance.

PEV charging can be divided into two types: 1) destination charging, which happens when a PEV arrives at its destination, including home charging, workplace charging etc. and 2) urgent charging, which happens when a PEV is still on the road and its SoC decreases to a certain threshold. Destination charging needs are mostly satisfied by distributed charging spots with low-power (or normal-power) charger deployed in private or public parking lots,[1] while urgent charging needs are mostly satisfied by FCSs. In recent years, both distributed charging spots and FCSs have gained heavy investments [9], [10]. But for the growing fleet of PEVs, their quantities are still not enough. Furthermore, due to the improper planning, the inconvenience for PEVs to get recharged and the under utilization of charging facilities occur simultaneously [10].

A. Literature Overview

Driven by the urgent needs of satisfying the increasing charging demands of PEVs and promoting the development of PEV industry, the planning for PEV charging facilities in urban areas has become a research focus for years.

[1]The distributed charging spots deployed in public parking lots mentioned here are also called public charging stations sometimes [7], [8].

Many researchers have studied on the optimal planning of FCSs in urban areas. Feng *et al.* [11] proposed to use the weighted Voronoi diagram to determine each charging station's location and service region. In [12], the optimal siting and sizing of FCSs was solved by the particle swarm optimization (PSO) algorithm. Bendiabdellah *et al.* [13] proposed to use an improved K-means clustering method to determine the service region of each charging station and solve the optimal siting problem using genetic algorithm. In [14], a two-step screening method was developed to locate the charging stations and a modified primal-dual interior point algorithm was proposed to determine their sizes. Lam *et al.* [15] proved the FCS placement problem to be NP-hard and proposed four different methods to solve it. In [16], a multiobjective electric vehicle (EV) charging station planning method which can ensure charging service while reducing power losses and voltage deviations of distribution systems was proposed. Yao *et al.* [17] studied the coordinated planning for the integrated power distribution network and PEV charging systems based on a decomposition-based multiobjective evolutionary algorithm. In [18], the technical design criteria for fast-charging infrastructure was studied. However, in [11]–[18], only the FCSs have been studied, while PLCSs were not considered.

Some researchers have studied the planning of PLCSs (stations). Marra *et al.* [7] studied the siting and sizing of public charging stations connected to the same distribution transformer, and compared the economic benefits between ac and dc charging stations. Gharbaoui *et al.* [8] developed an activity-based simulation tool to simulate EV drivers' behaviors to generate charging demands and studied the tradeoff between the number of public charging stations per parking area and the need of guaranteeing to satisfy PEV charging demands. However, in [7], the planning perspective is limited under a single transformer but not in a large urban area. In [7] and [8], only PLCSs (stations) were planned, while the integrated planning for different types of charging facilities was not considered.

To provide proper guidance for the deployment of PEV charging facilities, forecasts about the demands for different types of charging facilities is needed. In [7], [8], and [11]–[18], due to the lack of data or comprehensive work on PEV charging load forecasting, the charging demands were mainly estimated according to the traffic flow, which were assumed to be a Poisson distribution and so on, and the PEV penetration. However, since charging demands are influenced by various factors including the charging type, the arrival time, the parking duration, and so on, simply assuming the charging demands to be proportional to the traffic flow or PEV population is not adequate. In [18], though a mobility behavior simulation model was developed based on real-statistical transportation data to generate the charging demands of PEVs, the spatial distribution of charging demands was ignored.

Since different types of PEV charging facilities are substitutes for each other, it is essential to carry out an integrated planning, which is not considered in [7], [8], and [11]–[18]. In [19], though different types of charging facilities were

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TABLE I
MAIN CHARACTERISTICS OF DIFFERENT TYPES OF CHARGING FACILITIES

Type	Location	Customers	Charging Mode	Electricity Tariff	Planning Issues
HCS1	Residential private parking lots	Owners	Level 1/2	Residential tariff	–
HCS2	Residential public parking lots	Residents	Level 2	Residential tariff	Sizing
PCS1	Commercial/industrial public parking lots	Employees	Level 2	General commercial or industrial tariff	Sizing
PCS2	Commercial/Industrial public parking lots	The public	Level 2	General commercial or industrial tariff	Sizing
FCS	Roadside	The public	Level 3	Major industrial tariff	Sizing & Siting

planned together, the substitution effect between them was not studied.

B. Contributions of this Paper

In this paper, an integrated planning framework for different types of PEV charging facilities in urban area is proposed from the perspective of a social planner. The planning results can be used to provide guidance for the social planner to formulate PEV charging facility development plan.

Compared with the published papers on the similar topics, the major contributions of this paper can be summarized as follows.

First, different types of charging facilities including PLCSs deployed in public parking lots and roadside FCSs are studied and planned together. Furthermore, the substitution effect between the deployments of different types of charging facilities is studied. Thus, a minimization of the total social costs of the whole charging system can be achieved effectively.

Second, a method for forecasting the spatial and temporal distribution of PEV charging load is used to generate charging demands for different types of charging facilities as input data for the planning. This method simulates PEVs' driving, parking behaviors according to real-travel survey data and uses various factors including the charging type, the arrival time, the parking duration, and so on to determine PEVs' charging demands [20].

Lastly, the impacts of various factors, including the ambient temperature, the private charging spot possession rate and the service level (SL) of PLCSs, on the planning results are assessed. The simulation results indicate the impacts of these factors to be significant.

It should be noted that the charging facilities considered in this paper only provide charging services for private PEVs. Facility planning issues for public fleets (e.g., buses and taxies), the charging demands of which are strongly associated with their operating routines as studied in [22] and [23], is beyond the scope of this paper.

The remainder of this paper is organized as follows. The planning objective is introduced in Section II. The spatial and temporal distribution of PEV charging demand forecasting method is briefly introduced in Section III. In Section IV, the integrated planning framework is presented in detail. A case study on a real-urban area is carried out in Section V. Finally, Section VI concludes this paper.

II. PLANNING OBJECTIVE

A. Characteristics of Different Types of Charging Facilities

We divide charging facilities into five types, namely HCS1, HCS2, PCS1, PCS2, and FCS. Different types of charging facilities have different characteristics which will influence the PEVs' charging behaviors. Characteristics of these five types of charging facilities are compared in Table I. The electricity tariffs for different types of charging facilities may be different across countries or regions, which will influence PEVs' charging behaviors. The tariffs shown in Table I are introduced in [24], which are set by the Chinese government.

For economic reasons, it is not necessary to deploy charging spots for every parking lot. We define the SA index of HCS and PCS to represent the ratio of the number of charging spots to the peak PEV parking demands (PDs) of a certain type of parking lots. The PDs of different types of parking lots can be estimated by analyzing travel survey data.

For different types of charging facilities, the planning concerns are different. For the HCS2 and the PCS, since their locations are inside the public parking lots, only their capacities should be determined in the planning framework[2]; while for the FCS, their number, locations (sites), and capacities (sizes) should be determined. Since the HCS1 are personal property, their SA is not included in the planning variables but will serve as the input data.

B. Planning Objective

The objective of the integrated planning framework is to minimize the equivalent annual social costs of the whole PEV charging system, which can be expressed as follows:

$$\min F = C_I + C_{O\&M} + C_E + C_T. \tag{1}$$

The decision variables include the SAs of PLCSs (HCS2 and PCS) and the number of FCSs, I_F, and each FCS's site, l_{i_F}, and size, N_{i_F}.

C. Cost Analysis

To deploy different types of charging facilities, their incurred costs are different. Detailed comparison of costs for different types of charging facilities is shown in Table II.

1) Investment Costs: The annual investment costs own to the investments of all the charging facilities, which can be calculated as follows:

$$C_I = \frac{r(1+r)^{m_H}}{(1+r)^{m_H}-1} \times C_I^H + \frac{r(1+r)^{m_P}}{(1+r)^{m_P}-1} \times C_I^P + \frac{r(1+r)^{m_F}}{(1+r)^{m_F}-1} \times C_I^F \tag{2}$$

$$C_I^H = \left(C_C^H + C_G \times P^H\right) \times \mathrm{SA}^H \times \mathrm{PD}^H \tag{3}$$

[2]The planned number of PLCSs in a parking lot is the product of the SA value and the corresponding peak PD value.

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TABLE II
COSTS COMPARISON OF DIFFERENT TYPES OF CHARGING FACILITIES

Type	Investment Costs				Operation & Maintenance Costs	User Costs	
	Charging Facilities	Grid Reinforcement	Land Use	Buildings etc.		Electricity Costs	Time Costs
HCS1	√	–	–	–	√	√	–
HCS2	√	√	–	–	√	√	–
PCS1	√	√	–	–	√	√	–
PCS2	√	√	–	–	√	√	–
FCS	√	√	√	√	√	√	√

$$C_I^P = \left(C_C^P + C_G \times P^P\right) \times \mathrm{SA}^P \times \mathrm{PD}^P \tag{4}$$

$$C_I^F = \sum_{i_F=1}^{I_F} \left(\left(C_C^F + C_G \times P^F + C_L \times F_L^F\right) \times N_{i_F}^F + C_{IO}^F\right). \tag{5}$$

In (3)–(5), the first and the second parts are, respectively, the investment costs for the charging facilities and for the grid reinforcements. In (5), the third part in the parentheses is the land costs and the last part is other investment costs for FCSs such as costs for buildings and road improvements. For the HCS2 and PCS, the costs are approximately proportional to the total number of the charging spots. While for the FCS, since each FCS needs considerable investments on auxiliary facilities (C_{IO}^F), the costs for a FCS are not proportional to its capacity. Therefore, the investments costs for each FCS should be calculated separately in (5).

2) Operation and Maintenance Costs: The annual operation and maintenance costs can be calculated as follows:

$$C_{\mathrm{O\&M}} = C_{\mathrm{O\&M}}^H + C_{\mathrm{O\&M}}^P + C_{\mathrm{O\&M}}^F. \tag{6}$$

Since, it is hard to obtain detailed power network information for a city covering a large area in the long-term future to make a good estimation of $C_{\mathrm{O\&M}}$. We estimate it to be 10% of the investment costs [32].

3) Electricity Costs: Since different types of charging facilities are usually connected to the power network at different voltage levels, the corresponding per-unit electricity cost of PEV charging may be different. The annual electricity costs related to different types of charging facilities are calculated as follows:

$$C_E = C_E^H + C_E^P + C_E^F \tag{7}$$

$$C_E^H = \sum_{s\in S}\sum_{t\in T}\sum_{n_H=1}^{N_H} \left(D_s \times P_{n_H}(t) \times \Delta t \times \mathrm{EC}_H\right) \tag{8}$$

$$C_E^P = \sum_{s\in S}\sum_{t\in T}\sum_{n_P=1}^{N_P} \left(D_s \times P_{n_P}(t) \times \Delta t \times \mathrm{EC}_P\right) \tag{9}$$

$$C_E^F = \sum_{s\in S}\sum_{t\in T}\sum_{i_F=1}^{I_F}\sum_{n_{i_F}=1}^{N_{i_F}} \left(D_s \times P_{n_{i_F}}(t) \times \Delta t \times \mathrm{EC}_F\right). \tag{10}$$

The annual electricity costs related to the FCS (C_E^F) in (10) can also be calculated by (11), as follows:

$$C_E^F = \sum_{s\in S}\sum_{i_F=1}^{I_F}\sum_{j\in J_{i_F}} \left(D_s \times \frac{\left(\mathrm{SoC}_j^d - \mathrm{SoC}_j^a\right) \times B_j}{\eta} \times \mathrm{EC}_F\right). \tag{11}$$

Driving a PEV to its nearest FCS costs extra power, which will lead to the increase of total power consumption. So the arrival SoC of a PEV j (SoC_j^a) to a FCS is less than its charging threshold SoC ($\mathrm{SoC}_j^{\mathrm{thr}}$, the threshold when charging demand occurs), which can be calculated as follows:

$$\mathrm{SoC}_j^a = \mathrm{SoC}_j^{\mathrm{thr}} - \frac{d_{i_F j} \times E_j}{B_j}. \tag{12}$$

In (12), $d_{i_F j} \times E_j$ is the extra electricity consumed by driving PEV j to its nearest FCS. Thus, the corresponding extra electricity costs can be calculated as follows:

$$C_{E_{\mathrm{ex}}}^F = \sum_{s\in S}\sum_{i_F=1}^{I_F}\sum_{j\in J_{i_F}} \left(D_s \times \frac{d_{i_F j} \times E_j}{\eta} \times \mathrm{EC}_F\right). \tag{13}$$

4) Time Costs: Driving a PEV to its nearest FCS and waiting a PEV to be recharged both will cost the driver's extra time. The annual time costs can be calculated as follows:

$$C_T = C_T^F = \sum_{s\in S} D_s \times \left(\mathrm{TC}_{\mathrm{dr}} \times \sum_{i_F=1}^{I_F}\sum_{j\in J_{i_F}} \frac{d_{i_F j}}{v} + \mathrm{TC}_{\mathrm{ch}} \times \sum_{i_F=1}^{I_F}\sum_{j\in J_{i_F}} \frac{\left(\mathrm{SoC}_j^d - \mathrm{SoC}_j^a\right) \times B_j}{P^F \times \eta} + \mathrm{TC}_{\mathrm{ch}} \times \sum_{i_F=1}^{I_F}\sum_{j\in J_{i_F}} T_j^{\mathrm{w}}\right). \tag{14}$$

In (14), the first part in the parentheses is the time costs for driving the PEVs which need to get recharged to their nearest FCSs, which are proportional to the distances from the PEVs to the corresponding FCSs. The second part is the costs for waiting the PEVs to get recharged in the station and the third part is the costs associated with the queuing time. Since queuing or waiting a PEV to get recharged both do not need driver's extra work, their per-unit time costs are assumed to be the same ($\mathrm{TC}_{\mathrm{ch}}$). While driving a PEV to the nearest FCS need extra labor, the corresponding per-unit time cost ($\mathrm{TC}_{\mathrm{dr}}$) is higher.

The electricity and time costs in different types of days in one year are calculated separately, since the corresponding charging demands are different. The day type is determined by the ambient temperature in this paper, which has significant influence on PEVs' auxiliary power and will further influence their charging demands (see Section II-D for detailed information).

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D. Other Planning Considerations

1) Ambient Temperature: The ambient temperature has great influence on the driving range of PEVs [25]. Under extreme temperatures, the auxiliary loads (for heating or air conditioning etc.) increase significantly [26], which will lead to the decrease of PEVs' driving range and therefore the increase of charging demands.[3] The ambient temperature is the major factor to divide day types in this paper, and its impacts on the planning results are also assessed.

2) Private Charging Spot Possession Rate: Though the possession rate of private charging spots (the SA of HCS1) is beyond the scope of the planning framework, it influences the charging demands for other charging facilities. Before carrying out the planning for the other charging facilities, the SA of HCS1 should be estimated. This paper will also carry out a sensitivity analysis to investigate how the value of the SA impacts the planning results.

3) Public Charging Spot Service Level: The PEV drivers may not wait for their PEVs to be fully recharged before they leave the parking lots to complete their schedule such as going back home, working, or shopping. Thus, after a PEV has got fully recharged in a public parking lot, the charging spot it used may not be available for the others until the driver comes back. Compared with the FCS, the turnover rate of PLCSs may be lower. We define the SL index of PLCSs to represent the probability of a charging spot in use being given up for the others as soon as the PEV gets fully recharged. The impact of the SL of PLCSs on the planning results is also assessed.

III. Spatial and Temporal Distribution of PEV Charging Demands

Forecasting the spatial and temporal distribution of PEV charging demands is an essential preliminary work for PEV charging facility planning. The forecasting method proposed in [20] is used and introduced briefly in this section. Readers may need to refer to [20] for detailed information.

A. Input Data of the Forecasting Method

Since the penetration of PEVs is still very low and the PEV charging load recording data is quite limited, it is impractical to forecast PEV charging demands on the basis of recorded PEV charging load. The proposed method uses practical travel survey data [21] to simulate PEVs' driving and parking behaviors and uses a model to generate charging demands for different types of charging facilities.

The input data for the forecasting method includes the driving and parking behaviors of PEVs, the land development planning data, the PEV population and the SAs of different types of charging facilities in the studied area.

The driving and parking behaviors concerned in the forecasting method include: 1) the temporal probability distribution of PEV parking number; 2) the temporal probability distribution of the number of arrival PEVs; and 3) the probability distribution of PEV parking duration.[4] PEV drivers are more willing to charge their PEVs at their trip destination because it does not cost extra time. The arrival time and parking duration of PEVs determine their potential charging start time and duration. In the proposed method, parking places are divided into three types: 1) home, where drivers park their PEVs after going back home; 2) workplace, where drivers park their PEVs after arriving at workplace; and 3) other place, where drivers park their PEVs for other purposes. Parking and driving behaviors at these three types of places are all analyzed. The driving and parking behaviors of PEVs can be obtained by statistical travel survey data [20].

The spatial distribution of PEVs are highly influenced by the land development in urban area. For simplicity, blocks of different land usages are divided into three types: 1) residential areas where vehicles mostly park at home; 2) commercial areas where vehicles mostly park at other places or workplaces; and 3) industrial areas where vehicles mostly park at workplaces with a few park at other places. The land development planning data can be obtained from the land development planning documents issued by the government.

The PEV population and the SAs of different types of charging facilities (excluding the FCS) can be obtained from the PEV development planning documents issued by the government or via proper forecasting method.

It should be mentioned that the capacity of FCSs is excluded from the input data. We assume that all the charging demands for the FCS could be satisfied so that the number of FCSs and each FCS's capacity are determined by the charging demands for them, which are the outputs of the forecasting method.

B. Framework of the Forecasting Method

The framework of the PEV charging demand forecasting method is shown in Fig. 1.

First, the parking generation rate method [20] is used to calculate the spatial and temporal probability distribution of PEVs and the SAs of HCS and PCS are determined to obtain the deployments of HCS and PCS.

Second, the spatial and temporal probability distribution of PEV arrivals are calculated by combining the temporal probability distribution of the number of arrival PEVs and the spatial and temporal distribution of PEVs.

Third, a PEV charging demand generation model is built, which defines the time and the place the PEV should get recharged. Destination charging demand occurs at two circumstances.

1) A PEV arrives at its destination where there are charging spots available and its SoC has decreased to a threshold level.[5]

[3]Though the PEV battery may degrade under extreme temperatures [27], adequate cooling/heating systems are being used to keep the battery working under optimal operating temperature range [28].

[4]The probability distributions used here are obtained in [20] from the National Household Travel Survey data [21], in which, the arrival time, parking duration, and so on are surveyed down to the minute. However, for the convenience of simulation, we have changed them into discrete distributions approximately and the time intervals are all 15 min.

[5]The charging demand generation model uses various factors including the parking place, the arrival time, the parking duration, and so on to determine the SoC threshold [20].

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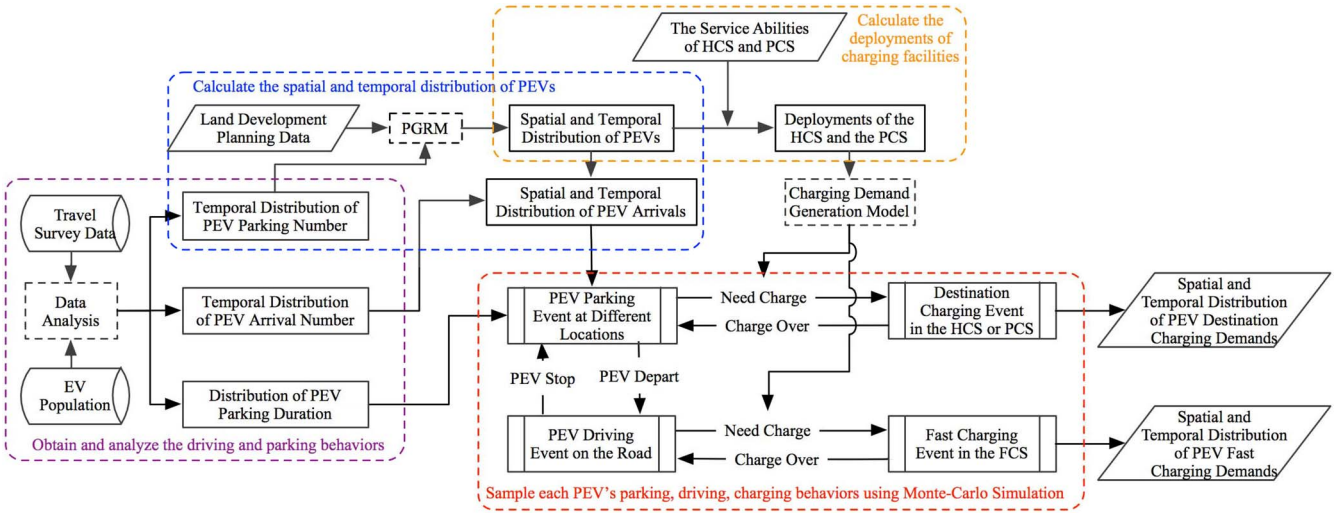


Fig. 1. Framework of the PEV charging demand forecasting method.

2) A PEV has finished its last trip in one day and been parked where there are charging spots available.

For the regular charging demand, the PEV will be charged at its destination. Urgent charging demand occurs when a PEV is still on the road while its SoC decreases under the security level. When urgent charging demand occurs, the PEV should be driven to the nearest FCS and get recharged promptly.[6]

Lastly, the Monte Carlo simulation method is used to sample each PEVs parking, driving and charging behaviors according to the spatial and temporal distribution of PEV arrivals, the distribution of PEV parking duration and the charging demand generation model. The simulation time interval is 15 min.

The outputs of this simulation include the spatial and temporal distribution of PEV charging loads and the charging demands for the FCS. Since the driving patterns vary with the days of the week, the charging demands are forecasted for a number of consecutive days and a whole week's results are selected for charging facility planning.

IV. Integrated Planning Framework for Different Types of PEV Charging Facilities

A. Integrated Planning Framework

The integrated planning framework can be briefly described in pseudo code in Table III. The core idea of the proposed framework is to decouple the planning for different types of charging facilities by traversing the SAs of HCS2 and PCS, which is based on the two points.

1) For the HCS2 and the PCS, only the numbers of their charging spots need to be planned, which are the products of their SA values (from 0% to 100%) and the corresponding peak PDs (see Section II).

[6]When urgent charging demands occur, there may be occasionally some PEV drivers tending to get their PEVs recharged in nearby public parking lots instead. However, it will cost much more extra time which may be unbearable for those who have not arrived at their destinations. So in this paper, this possibility is ignored.

TABLE III
Framework for the Integrated Planning

01	Set the private charging spot possession rate SA_{HCS1}
02	Calculate the maximum service abilities of HCS2 and PCS ($SA_{\text{HCS2}}^{\max}$, $SA_{\text{PCS}}^{\max}$) according to the distribution network power supply limits.
03	**for** $SA_{\text{HCS2}} = 0 : SA_{\text{step}} : SA_{\text{HCS2}}^{\max}$ **do**
04	**for** $SA_{\text{PCS}} = 0 : SA_{\text{step}} : SA_{\text{PCS}}^{\max 7}$ **do**
05	Use the method introduced in Section III to forecast the spatial and temporal distribution of charging demands for the HCS2, the PCS and the FCS.
06	Calculate the costs related to the HCS2 and the PCS.
07	Solve the optimal siting and sizing problem of FCSs and calculate the related costs (see Section IV-B).
08	Calculate the lowest equivalent annual social costs F related to the SA_{HCS2} and SA_{PCS}.
09	**endfor**
10	**endfor**
11	Output the lowest $F_{\min}$, the related $SA_{\text{HCS2}}, SA_{\text{PCS}}$, and the related site and size of each FCS.

2) The spatial and temporal distribution of PEV charging demands for the FCS with certain SAs of HCS2 and PCS can be forecasted by the method introduced in Section III.

To guarantee the planning accuracy of the framework, the SA_{step} is set to be a very small value. At step 04 in Table III, the charging demands for different types of charging facilities in all types of days are forecasted which are used to determine the capacity of each FCS.

B. Optimal Siting and Sizing of FCSs

Because the optimal siting and sizing of FCSs is a NP-hard problem [15]. It is solved by PSO algorithm, which does not use the gradient of the problem being optimized and can therefore be used on optimization problems that are partially irregular, noisy, change over time like the optimal siting and sizing problem of the FCS [29], [30]. Besides, Voronoi diagram [31] is employed to decide the service region of each FCS.

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TABLE IV
OPTIMAL SITING AND SIZING OF FCS

01	Input the spatial and temporal distribution of PEV charging demands for the FCS.
02	Set the range of the number of FCSs as $I_{F,\min}$ to $I_{F,\max}$ according to the coverage of the studied area and each FCS's maximum service range.
03	**for** $n = I_{F,\min}$ **to** $I_{F,\max}$ **do**
04	Initialize a quantity of particles' positions (all the n FCSs' sites) and initialize each particle's velocity.
05	**While** the termination criterion is not met **do**
06	Update each particle's position.
07	Make Voronoi diagram to decide each FCS's service region according to its site in every particle.
08	Calculate the optimal number of charging spots in each FCS according to the charging demands in every particle.
09	Calculate the C^{F} in every particle.
10	Update every particle's best known position and update the swarm's best known position.
11	Update every particle's velocity.
12	**endwhile**
13	**endfor**
14	Output the swarm's best known position, the related C^{F}, the number of FCSs and the corresponding capacity of each FCS.

In the PSO algorithm, a number of candidate solutions (the particles) are placed in the search space of the problem, and each evaluates the fitness function at its current location. Then each particle determines its movement through the search space by combining some aspect of its own current and best (best-fitness) history locations with those of the whole swarm. The next iteration takes place after all particles have been moved. Eventually, the swarm as a whole is likely to move close to an optimum of the fitness function [29].

In this paper, the candidate solutions are a series of randomly chosen locations of all the FCS. The fitness function is the equivalent annual social costs related to the FCS, which are calculated as follows:

$$C^F = C_I^F + C_{\mathrm{O\&M}}^F + C_E^F + C_T^F + \mathrm{PE} \tag{15}$$

where C_I^F, $C_{\mathrm{O\&M}}^F$, C_E^F, and C_T^F are the costs related to the FCS; PE is a penalty. When a candidate solution violates the constraints, PE will be set to be a large number; otherwise PE will be set to be 0. The constraints include the network limits,[7] the maximum drive range limit of PEVs, and so on.

Normally, a PEV tends to go to its nearest FCSs to get recharged, then a FCSs service region is approximately the planar ordinary Voronoi polygon associated with its location (l_{i_F}) ignoring the influence of traffic condition. The planar ordinary Voronoi polygon is defined as follows [31].

[7]For long-term planning, the power network information may be hard to obtain and for a large urban area, charging facilities at different sites usually will connect to different distribution networks and a large number of distribution networks will be involved. Thus, considering network limits for a large urban area may be feasible in theory, but it is quite difficult in practice at planning stage. In that case, network limits can be ignored. While at implementation stage, connecting the charging stations to the networks should take the network limits into account.

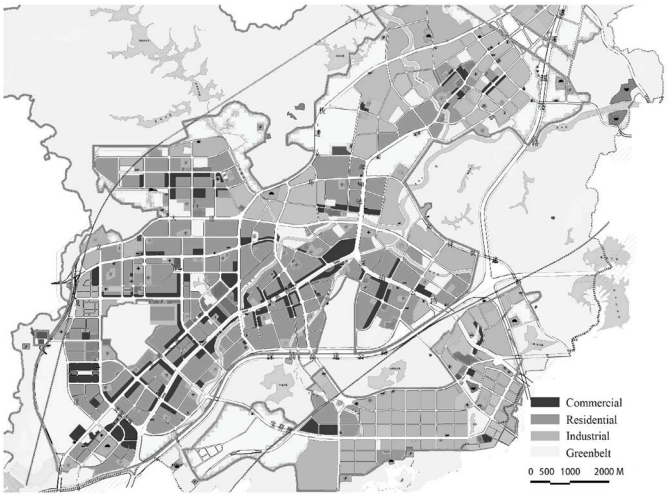


Fig. 2. Land development planning map of Longgang, 2020.

Let $L = [l_1, \ldots, l_{I_F}] \subset \mathbb{R}^2$ to represent the set of locations of all the FCS, where $2 < I_F < \infty$ and $l_{i_F} \neq l_{j_F}, i_F, j_F \in I_{I_F}$. We call the region given by

$$V(l_{i_F}) = \left\{ l \,\middle|\, \| l - l_{i_F} \| \leq \| l - l_{j_F} \| \quad \text{for } j_F \neq i_F, j_F \in I_{I_F} \right\} \tag{16}$$

the planar ordinary Voronoi polygon associated with l_{i_F}.

The set of service regions of all the FCS given by $\mathbb{V} = \{V(l_1), \ldots, V(l_{I_F})\}$ is the planar ordinary Voronoi diagram generated by L.

The proposed solution method of the optimal siting and sizing of FCS is briefly described in pseudo code in Table IV.

At step 08, some PEVs may have to wait for a while before getting recharged at peak hours. The capacity of each FCS is set to minimize the sum of the equivalent annual investment costs on the charging spots and the corresponding annual waiting time costs due to the lack of charging spots.

V. CASE STUDIES

A. Case Overview and Parameter Settings

The land development planning map of Longgang District in Shenzhen, China, 2020, is shown in Fig. 2 on which the blocks of different types of land usages are marked by different colors. This area covers an acreage about 196 km^2 with a population of 740 000 and a PEV population of 16 000 predicted in 2020. In this map, there are more than 400 blocks. We assume that the charging demands which arise in a block occur at its Voronoi center [31]. We also assume the numbers of PEVs coming into or out of this area are in dynamic equilibrium, and only the charging demands occur in this district are considered.

The parameters of charging facilities are shown in Table V, which are obtained from diverse sources.

In this paper, the HCS2 and PCS are assumed to be connected to 380 V distribution networks while the FCS are assumed to be connected to 10 kV distribution networks. The electricity costs are assumed to be the equal to the industrial tariffs under different voltage levels in Shenzhen [35].

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TABLE V
PARAMETERS OF THE CHARGING FACILITIES

Para.	Value	Para.	Value
r	0.08 [14] [32]	$m_{\mathrm{H/P/F}}$	10 [32]
$C_{\mathrm{C}}^{\mathrm{H}}$	\$2000 [33]	$C_{\mathrm{C}}^{\mathrm{P}}$	\$2000 [33]
$C_{\mathrm{C}}^{\mathrm{F}}$	\$23500 [33]	C_{G}	\$106.5 kVA^{-1} [14] [32] [33]
P^{H}	6.6 kW [33]	P^{P}	6.6 kW [33]
P^{F}	44 kW [33]	C_{L}	\$407 m^{-2} [34]
$F_{\mathrm{L}}^{\mathrm{F}}$	20 m^2 [12]	$C_{\mathrm{IO}i_{\mathrm{F}}}^{\mathrm{F}}$	\$163000 [14] [32] [33]
η	90% [11] [14]		

TABLE VI
PARAMETERS OF THE ELECTRICITY COSTS AND TIME COSTS

Para.	Value	Para.	Value
$EC_{\mathrm{H/P}}$	\$0.097 kWh^{-1} [35]	EC_{F}	\$0.094 kWh^{-1} [35]
TC_{ch}	\$5.76 h^{-1} [36]	TC_{dr}	\$8.22 h^{-1} [36]

A person's time costs are highly connected with his/her income level, activity purposes, and so on. In this paper, the time costs for PEV drivers are estimated according to a previous study on travel time costs [36]. Per-unit time cost for queuing or waiting PEVs to get recharged ($\mathrm{TC_{ch}}$) is set equal to 70% of the average hourly wages, while driving a PEV to the nearest FCS need extra labor, the corresponding per-unit time cost ($\mathrm{TC_{dr}}$) is set equal to 100% of the average hourly wages. The average wages is estimated based on that in Shenzhen, 2010 [37], which is targeted by the government to get doubled by 2020 [38].

The parameters of electricity costs and time costs are shown in Table VI.

For simulation analysis, the Nissan Leaf PEV is chosen to represent the whole PEV population,[8] whose battery capacity is 24 kWh and energy consumption is 0.14 kWh/km under 20 °C [26].

The extreme average temperature in Shenzhen is, respectively, about 6 °C and 35 °C in winter and in summer. We estimate that under 6 °C and 35 °C, the auxiliary power will increase by 0.6 and 0.4 kW, respectively, than that under 20 °C (1.6 kW) according to the data from [26]. For simplicity, only three types of days (e.g., days with an average temperature of 6 °C, 20 °C, and 35 °C) are considered.

The average driving speed (v) in this studied area is assumed to be 30 km/h and the departure SoC (SoC^d) is assumed to be 100%.

Six scenarios are simulated, which are compared in Table VII. In the scenario S0, the influence of extreme ambient temperature on the planning is ignored. The scenarios S1, S2, and S3 are simulated to assess the impact of private charging spot possession rate. The scenarios S4 and S5 are simulated to investigate the impact of the SL of PLCSs'.

B. Results Analysis

The summary of the planning results in all the six scenarios is shown in Table VIII.

[8]While doing practical planning, the planners should consider the mix of all types of PEVs in the targeted area, and investigate their parameters such as battery capacity, energy consumption, and so on. And then use the proposed forecasting method to obtain the charging demands of all the PEVs.

TABLE VII
SCENARIOS OF THE INTEGRATED PLANNING

Scenario	SA of HCS1	SL of HCS2 & PCS	Considering Extreme Temperature
S0	50%	50%	No
S1	30%	50%	Yes
S2	50%	50%	Yes
S3	80%	50%	Yes
S4	50%	00%	Yes
S5	50%	100%	Yes

From this table, we can observe that the ambient temperature, the private charging spot possession rate and the SL of PLCSs all have significant impacts on the planning results.

Comparing the scenario S0 with the scenario S2, if the influence of the extreme weather on the integrated planning is ignored, less charging facilities may be deployed, which may cause the shortage of charging service delivery in case of extreme climate conditions.

Comparing the scenarios S1 and S3 with the scenario S2, the higher the private charging spot possession rate is, the lower the SAs of PLCSs and the smaller the sizes of FCSs are. But, with lower private charging spot possession rate, the total quantity of charging facilities including the HCS, PCS and FCS is less. Encouraging new PEV owners to use PLCSs other than to build expensive private charging spots is more economical from the point of view of social welfare.

Comparing the scenarios S4 and S5 with the scenario S2, when the SL of PLCSs become higher, less investments both on the PLCSs and the FCS are needed, which leads to the decrease of the equivalent annual social costs. That is because when the SL of public charging spots becomes higher, the turn over rates of the PCS and the HCS2 both will increase. Appropriate measures should be encouraged to enhance the SL of PLCSs.

From Table VIII, we can also observe the number of FCSs in all of the six scenarios are the same. Building less but larger FCSs needs less charging spots (associated with the diversity factor of charging demands in different blocks, whose peak values may not occur at the same time) and less fixed investments, but will increase the PEV drivers' time costs and extra electricity costs. The planning results indicate that the saved investment costs by deploying larger FCSs is higher than the corresponding increased time and extra electricity costs. However, since the service range of each FCS is limited due to the limited drive range of PEVs, the number of FCSs should not violate the minimum limit $I_{F,\min}$. Thus, the optimal numbers of FCSs in all of the six scenarios are the same, while the total numbers of charging spots are different.

The planning results under the scenario S5 are selected to illustrate the detailed information about the integrated planning. The optimal distribution of all the FCSs in scenario S5 is shown in Fig. 3, in which the location of each FCS is marked by "Δ" and the service range of each FCS is defined by Voronoi diagram and the detailed information of all the FCSs in the scenario S5 is shown in Table IX.

The lowest annual social costs (F) and the annual costs related to the FCS (C^F) varying with the SAs of HCS2 and PCS under the scenario S5 are presented in Figs. 4 and 5

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TABLE VIII
SUMMERY OF THE INTEGRATED PLANNING RESULTS

Scenario	HCS2		PCS		FCS		Equivalent Annual Costs (million$)				
	SA	Spot Num.	SA	Spot Num.	Station Num.	Spot Num.	Investment	O&M	Electricity	Time	Total
S0	12.0%	1872	4.0%	320	12	79	1.602	1.075	3.746	1.378	7.801
S1	12.0%	1872	14.0%	1116	12	92	1.993	1.338	4.150	1.605	9.086
S2	10.0%	1560	9.0%	718	12	90	1.697	1.138	4.095	1.438	8.368
S3	5.0%	780	3.0%	240	12	82	1.146	0.769	4.063	1.173	7.151
S4	15.0%	2340	12.0%	957	12	93	2.123	1.425	4.131	1.589	9.268
S5	6.0%	936	6.0%	479	12	82	1.306	0.876	4.067	0.819	7.068

TABLE IX
PLANNING RESULTS OF FCSS

Station ID	01	02	03	04	05	06	07	08	09	10	11	12
X-coordinate	153	140	104	131	108	126	096	118	173	154	089	167
Y-coordinate	074	152	138	123	089	126	113	110	132	111	135	075
Number of Charging Spots	4	5	9	5	5	5	5	12	5	10	6	6

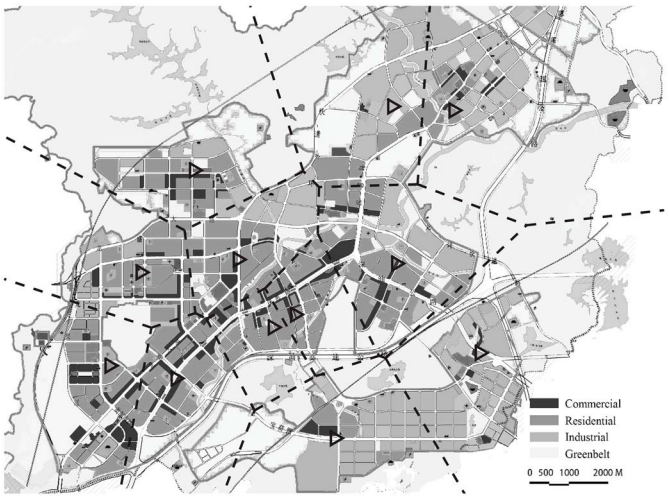


Fig. 3. Distribution of FCSs in Longgang, 2020.

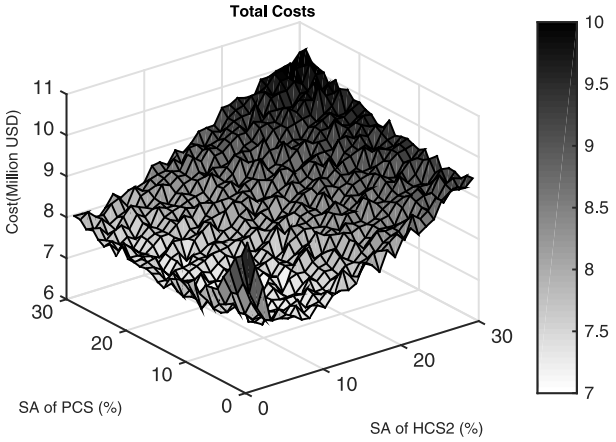


Fig. 4. Equivalent annual social costs.

to show the substitution effect between different types of charging facilities.

We can observe from Figs. 4 and 5 that there is a tradeoff between the deployments of PLCSs and the FCS. When the SAs of PLCSs is at a low level, with their increase, the charging demands for the FCS and the equivalent annual social costs

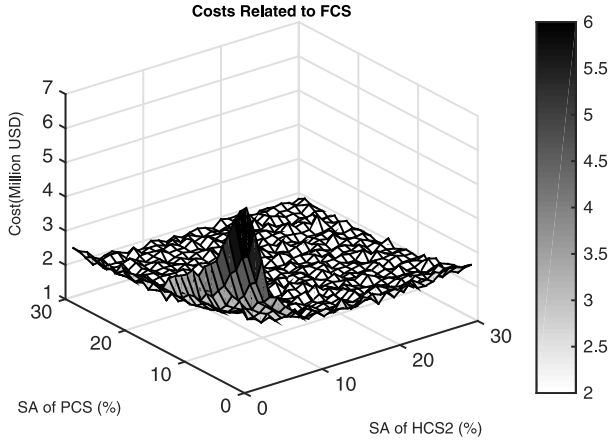


Fig. 5. Annual costs related to the FCS.

will decrease. But there is a diminishing return for the investments on PLCSs, which indicates that it is not economic to use PLCSs to replace FCSs completely since urgent charging demands happen inevitably when there are no PLCSs nearby or the drivers who have not arrived at their destinations are reluctant to wait too long to get their PEVs recharged.

C. Convergence Analysis

To guarantee the optimality of the planning results, the stability of the PSO algorithm should be ensured.

For the PSO algorithm shown in Table IV, the number of decision variables is $2\times I_F$ (the sites of all the FCS), population size is 50, initial weight factor is 0.9, final weight factor is 0.4, two acceleration constants are both 0.2. The convergence curves for C^F of 100 experiments in scenario S5, assuming that the SAs of the HCS1, HCS2, and PCS are, respectively, 50%, 6%, and 6%, are shown in Fig. 6 and the summary of the convergence results are listed in Table X.

The results show that the convergence of the PSO algorithm is stable and the worst objective value differs with the best objective value in less than 1% after 200 literation in 100 experiments, which has little influence on the planning.[9]

[9]Compared with Figs. 4 and 5, we can see that the disturbances caused by the PSO algorithm are negligible.

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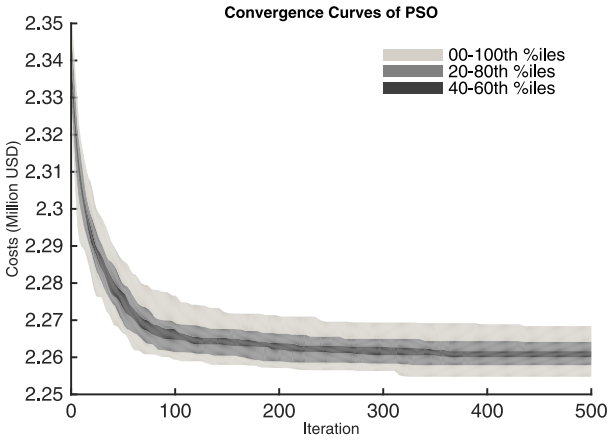


Fig. 6. Key percentiles of PSO convergence curves.

TABLE X
CONVERGENCE RESULTS OF PSO

Iteration	Best Object	Worst Object	Average Object
200	2.2549	2.2717	2.2630
500	2.2533	2.2710	2.2612

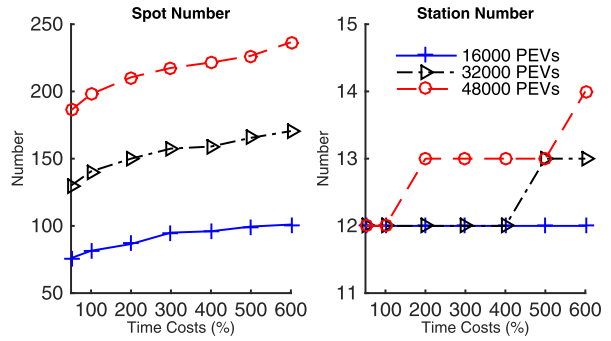


Fig. 7. Spot and station number of FCS under different time costs and PEV population.

D. Sensitivity Analysis

Since diverse resources are referenced to obtain the various parameters in the above simulations, further analysis on the sensitivities of some important parameters are provided.

1) Per-Unit Time Costs and PEV Population: With the increase of personal income, people will more value quality of life and the corresponding per-unit time costs for urgent charging will increase, which may influence the planning of FCS in the future. On the other hand, when PEV population grows, the utilization of the FCS will increase and thus the planning results of the FCS will also change.

Assuming that the SAs of the HCS1, HCS2, and PCS are, respectively, 50%, 6%, and 6%. The planning results for the FCS under different PEV population levels assuming the per-unit time costs being 50%–600% of the values (TC_{ch}, TC_{dr}) given in Table VI are shown in Fig. 7.

It can be seen that with the increasing of time costs, the total capacity of all the FCSs increases apparently. However, under low PEV population level, since the urgent charging demands are not very much, the number of FCS does not increase with the time costs. While under high PEV population level, with the increasing of time costs, the number of FCS increases, too. That is because when the per-unit time costs and the utilization of FCSs are high enough, it will be more economical to build more FCSs to reduce the drivers' time costs.

TABLE XI
PLANNING RESULTS OF THE FCS WITH DIFFERENT DEPARTURE SOCS

SOC^{d}	spot num.	Equivalent Annual Costs of FCS (million$) Invest.	O&M	Time	Electricity	Total
100%	82	0.736	0.494	0.702	0.329	2.261
60%	56	0.595	0.399	0.424	0.166	1.583

2) Departure SoC of Urgent Charging: The departure SoC (SoC^{d}) in the above simulations is assumed to be 100%. For destination charging demands, of which the parking time is usually long enough for PEVs to get fully recharged, setting SoC^{d} equal to 100% is reasonable. However, for urgent charging demands, due to the higher charging costs, drivers may charge their PEVs to less than 100% SoC before they leave the station. Assuming that the SAs of the HCS1, HCS2, and PCS is, respectively, 50%, 6%, and 6%, the planning results of FCS with the SoC^{d} to be 100% and 60% in scenario S5 are compared in Table XI.

From Table XI, we can see that choosing SoC^{d} to be 100% will make the planning results much more conservative. When doing practical planning, the distribution of SoC^{d} should be based on real-PEV charging survey data.

VI. CONCLUSION

There is a tradeoff between the deployments of PLCSs and FCSs, since they are substitutes for each other. This paper proposes an integrated planning framework for them in urban area, from the perspective of a social planner. Case studies on a real-urban area in China indicate its effectiveness. The simulation results show that the ambient temperature, the private charging spot possession rate and the SL of PLCSs' of the target area all have significant influence on the planning results. To avoid the shortage of charging service delivery in case of extreme climate conditions, the influence of extreme ambient temperatures in the target area should be carefully taken into account. Encouraging PEV owners to use PLCSs other than to build expensive private charging spots, taking measures to improve the SL of PLCSs both will help to save the equivalent annual social costs of the whole PEV charging system.

Study on integrated planning of different types of PEV charging facilities considering network and traffic conditions and adoption of increasing renewable resources will be our future work.

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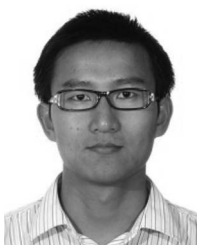
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